

FOLIOCRAFT — PROJECT PIPELINE & FEATURE SPEC

Comprehensive Technical Architecture, Feature Audit & Deployment Roadmap

PROJECT STATUS: PRODUCTION-READY • TARGET: WEB / MOBILE / ATS RESUME

1. EXECUTIVE OVERVIEW & SCOPE

MODULE 01

FolioCraft is an all-in-one developer & creative portfolio engine paired with an automated ATS resume generation studio. Built with modern TypeScript and React 19, the platform solves the disconnect between having a visually captivating online portfolio and generating a compliant, recruiter-friendly single-page PDF resume.

- **Target Audience:** Software engineers, UI/UX designers, engineering managers, students, and freelancers.
- **Primary Value Proposition:** Single source of truth: enter your career information once, and immediately get both a high-fidelity interactive web portfolio AND a pixel-accurate ATS-compliant PDF resume.
- **Zero Lock-in:** Full exportability via standalone self-contained HTML (zero CDN/server dependencies), raw JSON schema, and vector PDF.

2. TECHNICAL PIPELINE & DATA ARCHITECTURE

MODULE 02

The application is structured into a unidirectional data flow architecture designed for performance, resilience, and offline capability:

- **1. Centralized State Engine (App.tsx):** Manages the root PortfolioData state with automatic LocalStorage synchronization (key: foliocraft_portfolio_v2), ensuring continuous persistence during editing sessions without data loss.
- **2. Reactive Dual-Pipeline Renderers:** State updates feed in real-time into two independent rendering engines: (a) PortfolioRenderer for responsive web displays and (b) ResumeView for physical print/paper layouts.
- **3. CSS Color Module Level 4 Normalization:** Integrated with html2canvas-pro to support modern Tailwind v4 oklch colors natively, preventing canvas rendering crash exceptions during PDF capture.
- **4. Multi-Page Fitting Algorithm:** Calculates millimeter-precise document height against standard A4/US Letter dimensions (1050px standard page break height) with dynamic scale factors (0.80x to 1.05x).

3. DETAILED FEATURE INVENTORY & SPECIFICATIONS

MODULE 03

- **Feature A: Clean Canvas vs Archetype Presets:** Defaults to an unpolluted blank template ('(Blank C production deployments are ready for immediate use. Users can also load 4 curated archetypes: Product Designer, Fullstack Engineer, Art Director, and OSS Contributor.
- **Feature B: Live Split-Screen & Device Simulator:** Provides instant side-by-side editing on desktops ($\geq 1024\text{px}$) with switchable device simulation for Desktop (1440px), iPad/Tablet (768px), and Smartphone (375px) with simulated browser chomes and status notches.
- **Feature C: Mobile Bottom Navigation Bar (Phones/Tablets):** A dedicated thumb-friendly 5-action bottom bar (Edit, Preview, Resume, Themes, More) designed for touch screens, with an animated drawer for secondary tools.
- **Feature D: ATS Resume Studio with 4 Visual Templates:** Interactive template gallery offering Modern Minimalist, Tech Architect (2-column density), Harvard Classic (Ivy League serif

standard), and Executive Modern. Includes page-boundary guides, ATS plain-text copy, and 1-click PDF download.

- **Feature E: Key Numbers / Metrics Demystified:** Transformed corporate metric jargon into intuitive highlight numbers (e.g. 3+ Years Experience, 15+ Projects) with integrated onboarding guidance and toggleable visibility.
- **Feature F: Standalone Single-File HTML Export:** Generates a fully self-contained index.html with inlined CSS, SVGs, and responsive behaviors, hostable anywhere without Node.js or build steps.
- **Feature G: Social SEO & OpenGraph Meta Studio:** Simulates real-time social share preview cards for Twitter/X and LinkedIn with meta tag generation and verification.
- **Feature H: Unified 1-Click Export & Backup Hub:** Consolidated all download and export operations into a single non-redundant modal providing standalone HTML bundle, direct ATS resume PDF access, JSON backup/restore, and documentation without visual clutter.

4. TECHNOLOGY STACK & DEPENDENCY MATRIX

MODULE 04

LAYER / TOOL	TECHNOLOGY	ROLE & FUNCTION
Core Framework	React 19 + TypeScript	Reactive state, component architecture, type safety
Build Engine	Vite 6	Fast HMR, optimized production asset bundling
Design System	Tailwind CSS v4	Utility-first modern styling with CSS nesting & OKLCH
Iconography	Lucide React	Consistent, tree-shakable SVG icon suite
PDF Core Engine	html2canvas-pro + jsPDF	DOM-to-canvas rendering with modern CSS color support
Animation	Motion (Framer)	Smooth layout transitions & micro-interactions

5. ENGINEERING MILESTONES & GIT RELEASE LOG

MODULE 05

- **v1.0 (Core Engine):** Portfolio data models, Editor panel, Theme engine, and local storage cache.
- **v1.2 (ATS Resume):** Multi-template resume architecture with target 1-page/2-page scaling algorithm.
- **v1.5 (PDF Hardening):** Upgraded rendering pipeline to html2canvas-pro to eliminate color space crashes.
- **v1.8 (Clean Slate & UX):** Default clean blank template, simplified metrics explanations, and 1-click canvas reset.
- **v2.0 (Universal Responsive):** Full phone & desktop responsiveness, thumb-nav mobile dock, and professional docs.

VERIFICATION & DEPLOYMENT READINESS

All features in this specification have been validated through end-to-end linting, unit compilation, and responsive display checks. The repository is ready for live public deployment on Vercel, Netlify, or custom hosting.